Score Forecasts

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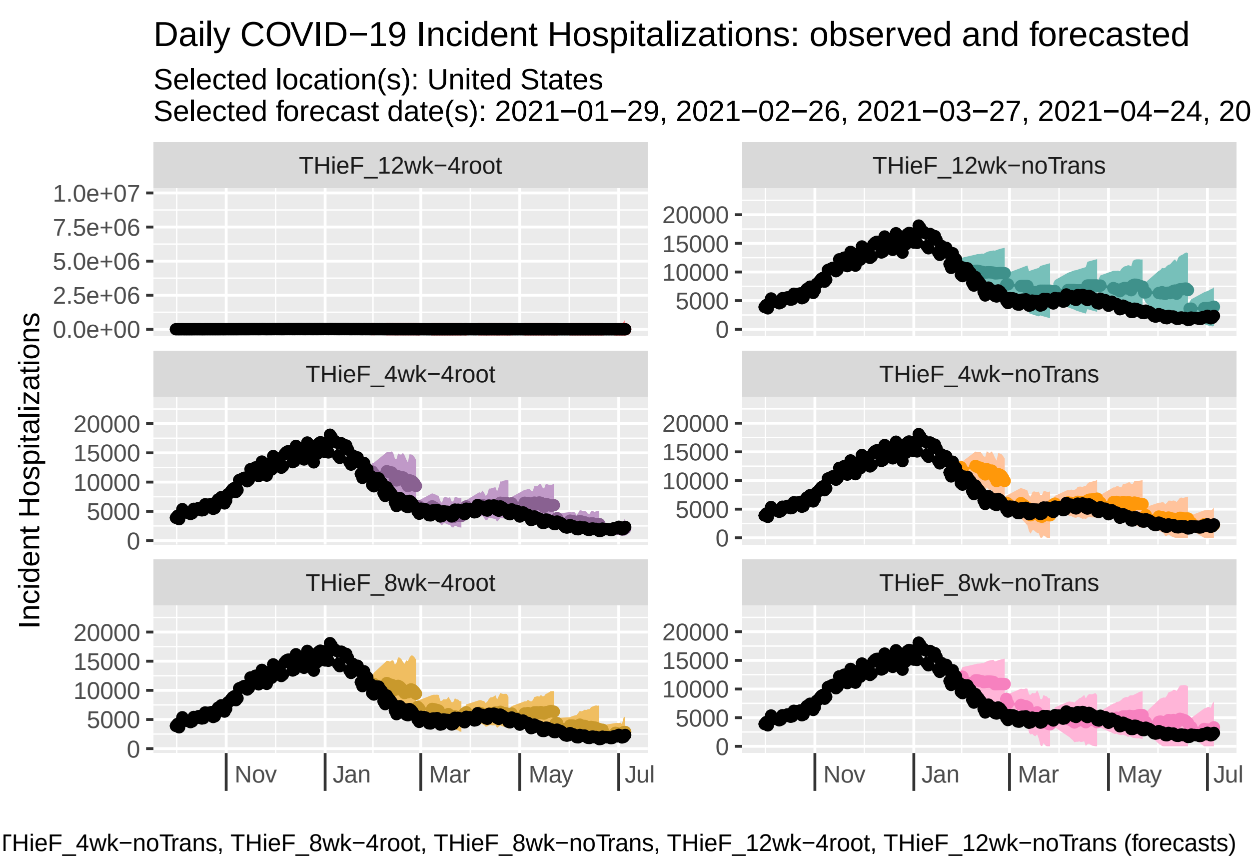
Load in truth data and forecasts

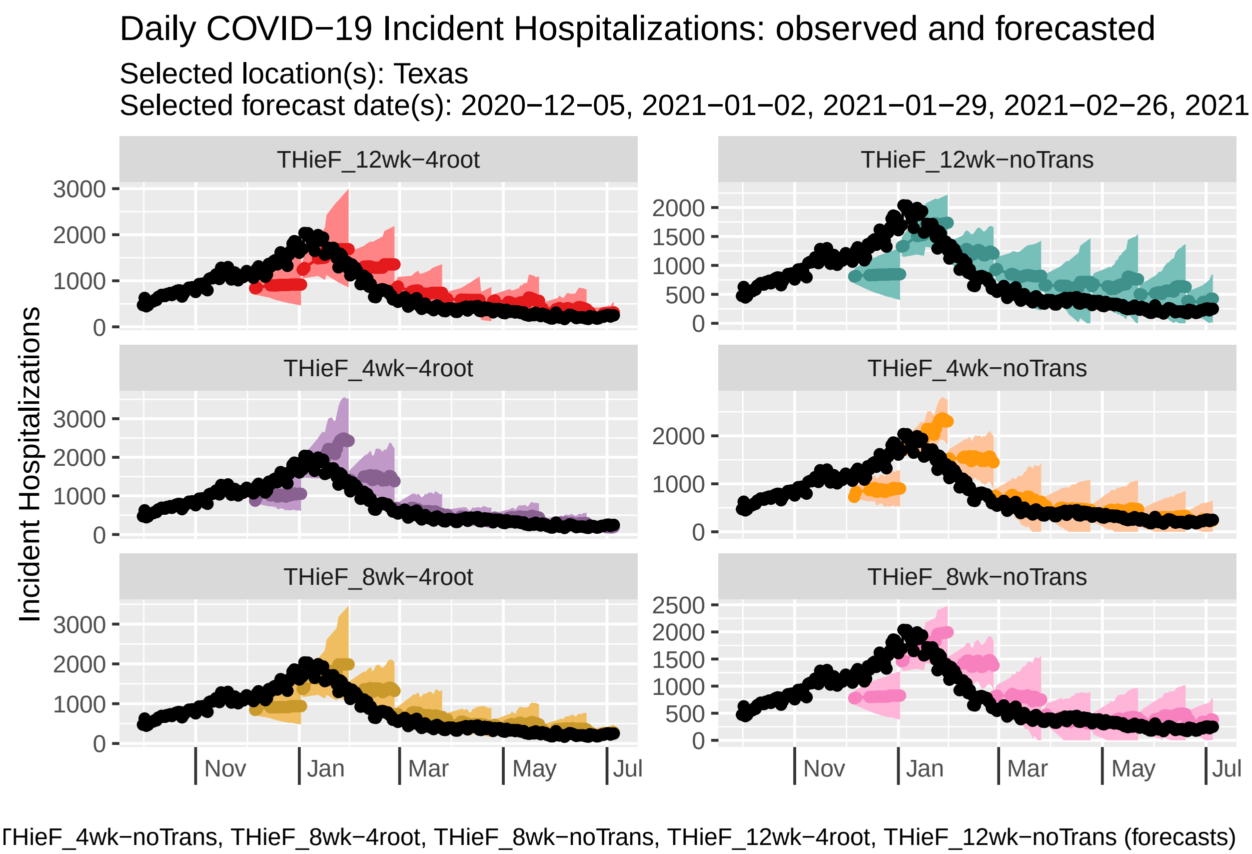
We load in the hospitalization truth data and our forecasts. (Note: The load forecast parentheses function wasn't working, so alternate code is written to achieve the same output.)

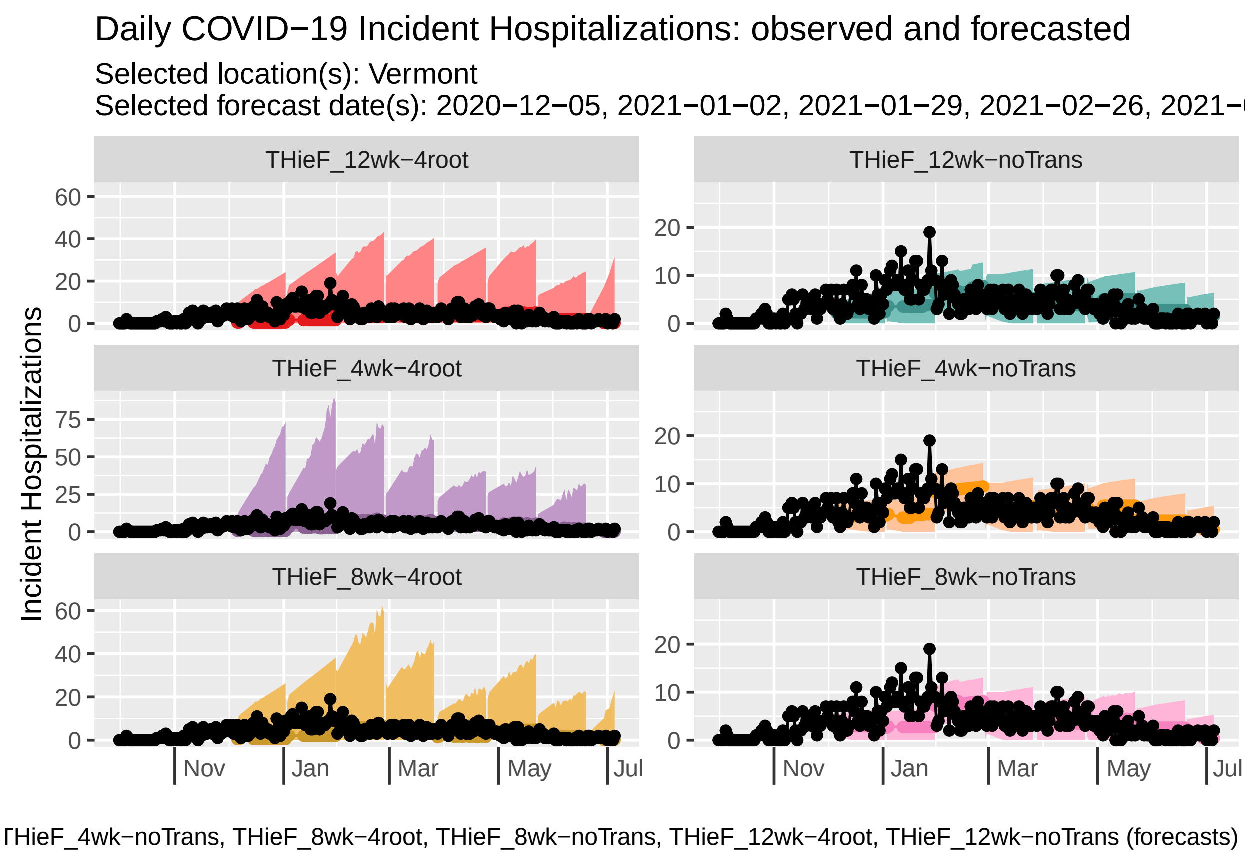
Versioned Truth Results

Plot Forecasts

We plot forecasts for three locations (US National, Texas, and Vermont) for each of the three models. The two state locations were chosen as the highest and lowest hospitalization count locations. The plotted forecasts are spaced 4 weeks apart with 1 - 28 day ahead horizons, starting with the first forecast date, 12/07/20, until the end of the training period, 07/05/21.




Daily COVID−19 Incident Hospitalizations: observed and forecasted
Selected location(s): Texas
Selected forecast date(s): 2020−12−05, 2021−01−02, 2021−01−29, 2021−02−26, 2021
THieF_12wk−4root
THieF_12wk−noTrans
THieF_4wk−4root
THieF_4wk−noTrans
THieF_8wk−4root
THieF_8wk−noTrans
Incident Hospitalizations
0
500
1000
1500
2000
2500
3000
Nov
Jan
Mar
May
Jul
THieF_4wk−noTrans, THieF_8wk−4root, THieF_8wk−noTrans, THieF_12wk−4root, THieF_12wk−noTrans (forecasts)



The plots don't show much difference between the 6 models (although this is partially due to the fact that the yscales are free and are not consistent even for the same location). However, of note is the fact that the models that use a forth root transformation tend to have wider prediction intervals than the models without transformations. All of the models reasonably follow the trend of the truth lines for every location shown, although it seems like lower count locations may have slightly greater accuracy. The baseline seems to have intervals of fairly consistent widths while the models tend to vary in their interval widths, tending to show wider intervals during times of higher incidents or change points and narrow lower intervals otherwise. We investigate model accuracy more deeply by scoring the forecasts for each model and calculating appropriate metrics.

Score forecasts

We calculate metrics for all three models, aggregating by horizon. Since hospitalizations are forecasted at a daily time scale, a new horizon week variable is created in which scoring metrics are averaged over from a single week Tuesday to the following Monday. We perform these calculations for the entire training phase as well as for each of the two waves that occurred during the training phase.

Overall metrics

model	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
COVIDhub-baseline	1093.262	1581.078	0.816	1.000	1.000	1.000	1.000	1.000
THieF_4wk-4root	1100.767	1643.299	0.278	0.705	1.007	1.039	-0.703	-4.905
THieF_8wk-4root	1178.802	1803.262	0.186	0.638	1.078	1.141	-0.994	-6.238

model	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
THieF_8wk-noTrans	1208.598	1887.624	0.376	0.859	1.105	1.194	-0.392	-1.825
THieF_4wk-noTrans	1232.919	1835.426	0.363	0.854	1.128	1.161	-0.432	-1.921
THieF_12wk-4root	1655.298	2336.802	0.151	0.516	1.514	1.478	-1.105	-8.683
THieF_12wk-noTrans	1665.758	2675.296	0.151	0.786	1.524	1.692	-1.105	-3.286

model	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
COVIDhub-baseline	31.950	41.770	0.889	0.991	1.000	1.000	1.000	1.000
THieF_4wk-4root	32.484	49.760	0.446	0.903	1.017	1.191	-0.140	-1.164
THieF_8wk-4root	33.213	51.024	0.397	0.881	1.040	1.222	-0.266	-1.692
THieF_4wk-noTrans	34.182	51.030	0.557	0.908	1.070	1.222	0.146	-1.020
THieF_8wk-noTrans	34.392	51.955	0.519	0.913	1.076	1.244	0.050	-0.906
THieF_12wk-noTrans	39.659	62.356	0.376	0.910	1.241	1.493	-0.319	-0.993
THieF_12wk-4root	102.494	56.907	0.365	0.841	3.208	1.362	-0.346	-2.663

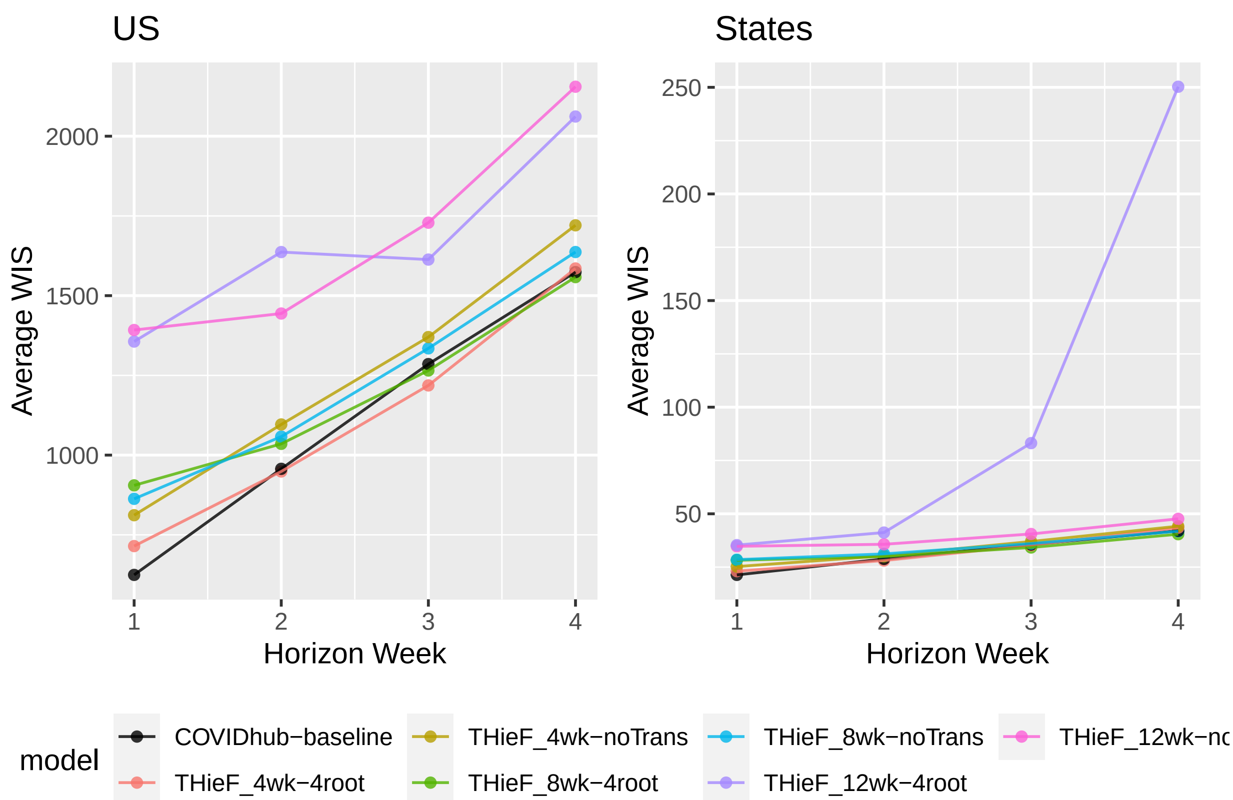
Horizon week metrics

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
COVIDhub-baseline	1	624.322	821.571	0.857	1.000	1.000	1.000	1.000	1.000
THieF_4wk-4root	1	714.668	1105.731	0.238	0.720	1.145	1.346	-0.733	-4.595
THieF_4wk-noTrans	1	811.526	1252.456	0.292	0.839	1.300	1.524	-0.583	-2.214
THieF_8wk-noTrans	1	862.786	1369.714	0.280	0.833	1.382	1.667	-0.617	-2.333
THieF_8wk-4root	1	905.091	1371.311	0.089	0.548	1.450	1.669	-1.150	-8.048
THieF_12wk-4root	1	1356.090	1901.621	0.101	0.476	2.172	2.315	-1.117	-9.476
THieF_12wk-noTrans	1	1392.227	2164.065	0.071	0.679	2.230	2.634	-1.200	-5.429
THieF_4wk-4root	2	948.675	1399.012	0.379	0.739	0.991	1.042	-0.345	-4.217
COVIDhub-baseline	2	956.969	1343.107	0.851	1.000	1.000	1.000	1.000	1.000
THieF_8wk-4root	2	1035.297	1617.233	0.199	0.652	1.082	1.204	-0.858	-5.957
THieF_8wk-noTrans	2	1057.921	1713.428	0.391	0.894	1.105	1.276	-0.310	-1.112
THieF_4wk-noTrans	2	1096.041	1571.374	0.429	0.863	1.145	1.170	-0.203	-1.733
THieF_12wk-noTrans	2	1443.980	2391.422	0.099	0.832	1.509	1.781	-1.141	-2.354

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
THieF_12wk-4root	2	1636.724	2066.041	0.174	0.596	1.710	1.538	-0.929	-7.075
THieF_4wk-4root	3	1218.497	1810.366	0.286	0.708	0.948	0.955	-0.726	-4.844
THieF_8wk-4root	3	1265.293	1942.346	0.227	0.675	0.984	1.024	-0.924	-5.494
COVIDhub-baseline	3	1285.460	1895.932	0.795	1.000	1.000	1.000	1.000	1.000
THieF_8wk-noTrans	3	1334.630	2058.841	0.442	0.864	1.038	1.086	-0.198	-1.727
THieF_4wk-noTrans	3	1370.170	2007.636	0.429	0.864	1.066	1.059	-0.242	-1.727
THieF_12wk-4root	3	1613.163	2417.267	0.149	0.558	1.255	1.275	-1.189	-7.831
THieF_12wk-noTrans	3	1728.802	2818.531	0.201	0.864	1.345	1.487	-1.012	-1.727
THieF_8wk-4root	4	1558.179	2354.961	0.238	0.687	0.990	0.992	-1.034	-5.259
COVIDhub-baseline	4	1573.898	2374.591	0.753	1.000	1.000	1.000	1.000	1.000
THieF_4wk-4root	4	1585.263	2350.191	0.204	0.646	1.007	0.990	-1.168	-6.075
THieF_8wk-noTrans	4	1636.803	2490.935	0.401	0.844	1.040	1.049	-0.389	-2.129
THieF_4wk-noTrans	4	1720.636	2610.469	0.306	0.850	1.093	1.099	-0.766	-1.993
THieF_12wk-4root	4	2061.735	3046.403	0.184	0.429	1.310	1.283	-1.249	-10.429
THieF_12wk-noTrans	4	2155.218	3420.416	0.245	0.776	1.369	1.440	-1.007	-3.490

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
COVIDhub-baseline	1	21.348	27.015	0.866	0.985	1.000	1.000	1.000	1.000
THieF_4wk-4root	1	23.090	35.089	0.447	0.898	1.082	1.299	-0.144	-1.476
THieF_4wk-noTrans	1	25.239	37.528	0.513	0.907	1.182	1.389	0.036	-1.212
THieF_8wk-4root	1	28.215	41.430	0.378	0.831	1.322	1.534	-0.334	-3.366
THieF_8wk-noTrans	1	28.497	41.488	0.480	0.892	1.335	1.536	-0.056	-1.632
THieF_12wk-noTrans	1	34.761	51.629	0.327	0.858	1.628	1.911	-0.474	-2.600
THieF_12wk-4root	1	35.323	50.002	0.325	0.775	1.655	1.851	-0.478	-4.936
THieF_4wk-4root	2	28.087	43.565	0.454	0.915	0.974	1.200	-0.116	-0.813
COVIDhub-baseline	2	28.828	36.312	0.899	0.993	1.000	1.000	1.000	1.000

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
THieF_8wk-4root	2	29.990	46.323	0.397	0.890	1.040	1.276	-0.258	-1.398
THieF_4wk-noTrans	2	30.305	45.568	0.558	0.915	1.051	1.255	0.144	-0.820
THieF_8wk-noTrans	2	31.185	47.115	0.516	0.922	1.082	1.298	0.039	-0.654
THieF_12wk-noTrans	2	35.698	56.255	0.383	0.923	1.238	1.549	-0.292	-0.643
THieF_12wk-4root	2	41.211	52.079	0.369	0.853	1.430	1.434	-0.328	-2.290
THieF_8wk-4root	3	34.211	53.730	0.403	0.906	0.961	1.151	-0.244	-1.025
THieF_4wk-4root	3	35.258	54.480	0.449	0.907	0.990	1.167	-0.128	-0.986
COVIDhub-baseline	3	35.599	46.682	0.899	0.993	1.000	1.000	1.000	1.000
THieF_8wk-noTrans	3	36.049	55.216	0.532	0.921	1.013	1.183	0.080	-0.684
THieF_4wk-noTrans	3	37.016	55.250	0.580	0.908	1.040	1.184	0.201	-0.975
THieF_12wk-noTrans	3	40.533	65.139	0.396	0.930	1.139	1.395	-0.260	-0.460
THieF_12wk-4root	3	83.152	58.056	0.384	0.873	2.336	1.244	-0.291	-1.779
THieF_8wk-4root	4	40.434	62.611	0.409	0.897	0.962	1.097	-0.233	-1.262
THieF_8wk-noTrans	4	41.836	64.000	0.550	0.917	0.995	1.121	0.127	-0.778
COVIDhub-baseline	4	42.026	57.073	0.892	0.992	1.000	1.000	1.000	1.000
THieF_4wk-4root	4	43.501	65.904	0.432	0.890	1.035	1.155	-0.173	-1.439
THieF_4wk-noTrans	4	44.168	65.772	0.576	0.904	1.051	1.152	0.195	-1.108
THieF_12wk-noTrans	4	47.644	76.402	0.398	0.927	1.134	1.339	-0.262	-0.541
THieF_12wk-4root	4	250.290	67.493	0.383	0.865	5.956	1.183	-0.299	-2.033



When splitting the results so that US National is separate, we find that for US National the two models that use the base set hierarchy tended to perform the best, with improvements seen over a naive baseline in all metrics except for 95%PI Coverage. These two models are followed by the Topmost8_arima_noTrans model, which showed more comparable performance to the baseline. The Base_arima_noTrans model performed the best in terms of PI Coverage, followed by the Base_arima_4root and Topmost8_arima_noTrans models (with the latter being more consistent with coverage rates for all horizons). In terms of WIS and MAE, the Base_arima_4root model performed the best, followed by the Base_arima_noTrans model. Overall, for the US National level, it seems that there is a pattern that for coverage rates, the models without transformations perform better (and these tend to have smaller prediction intervals) while for WIS and MAE the fourth root transform models perform better; these patterns hold at all horizons. Further, there is a general order to model accuracy in terms of hierarchical structure used: the Base set performs the best, followed by the Topmost8 set, and finally the Multiple3 set.

For the state level, we see noticeably better coverage rates that approach the nominal level (or even exceed it) for all of the models. (We also see shrunken WIS and MAE values, although this is likely partially due to the fact that the states have smaller scale values than the National level.) Interestingly, we see fewer improvements to MAE over the baseline but more substantial improvements in terms of coverage rates. It's a little harder to definitively say which models have the best coverage rates, as the results differ between the 50% and 95% intervals and the models are quite similar in coverage rates, especially for the 95% intervals. However, it seems that the Base_arima_4root model has the best performance for all metrics across all horizons. It is followed closely by the Topmost8_arima_noTrans and Base_arima_noTrans models, then the Topmost8_arima_4root and Multiple3_arima_noTrans models, all of which are much better than the Multiple3_arima_4root model.

Metrics by wave

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
alpha	THieF_4wk-4root	1	496.782	763.286	0.295	0.733	0.994	1.433	-0.434	-4.333
alpha	COVIDhub-baseline	1	499.666	532.705	0.971	1.000	1.000	1.000	1.000	1.000
alpha	THieF_4wk-noTrans	1	541.735	909.443	0.314	0.952	1.084	1.707	-0.394	0.048
alpha	THieF_8wk-noTrans	1	567.181	975.878	0.362	1.000	1.135	1.832	-0.293	1.000
alpha	THieF_8wk-4root	1	774.568	1120.501	0.095	0.495	1.550	2.103	-0.859	-9.095
alpha	THieF_12wk-4root	1	1418.600	1881.176	0.086	0.419	2.839	3.531	-0.879	-10.619
alpha	THieF_12wk-noTrans	1	1656.688	2509.296	0.010	0.543	3.316	4.710	-1.040	-8.143
alpha	THieF_4wk-4root	2	638.495	979.760	0.347	0.786	0.869	1.160	-0.312	-3.286
alpha	THieF_4wk-noTrans	2	718.497	1184.253	0.388	1.000	0.977	1.402	-0.229	1.000
alpha	THieF_8wk-noTrans	2	720.124	1230.278	0.520	1.000	0.980	1.456	0.042	1.000
alpha	COVIDhub-baseline	2	735.043	844.888	0.990	1.000	1.000	1.000	1.000	1.000
alpha	THieF_8wk-4root	2	852.160	1278.392	0.245	0.551	1.159	1.513	-0.521	-7.980
alpha	THieF_12wk-noTrans	2	1685.260	2749.065	0.041	0.776	2.293	3.254	-0.938	-3.490
alpha	THieF_12wk-4root	2	1818.032	1978.028	0.224	0.480	2.473	2.341	-0.562	-9.408
alpha	THieF_4wk-4root	3	841.435	1305.210	0.308	0.758	0.902	1.116	-0.393	-3.835
alpha	THieF_4wk-noTrans	3	910.769	1511.031	0.374	1.000	0.977	1.292	-0.258	1.000
alpha	COVIDhub-baseline	3	932.443	1169.429	0.989	1.000	1.000	1.000	1.000	1.000
alpha	THieF_8wk-noTrans	3	949.111	1600.001	0.451	1.000	1.018	1.368	-0.101	1.000
alpha	THieF_8wk-4root	3	1042.361	1610.995	0.176	0.626	1.118	1.378	-0.663	-6.473
alpha	THieF_12wk-4root	3	1627.555	2367.920	0.121	0.429	1.745	2.025	-0.775	-10.429
alpha	THieF_12wk-noTrans	3	1965.772	3239.286	0.110	0.846	2.108	2.770	-0.798	-2.077
alpha	COVIDhub-baseline	4	1113.129	1492.357	0.976	1.000	1.000	1.000	1.000	1.000
alpha	THieF_8wk-noTrans	4	1146.598	1918.433	0.417	1.000	1.030	1.286	-0.175	1.000
alpha	THieF_4wk-4root	4	1212.028	1845.041	0.214	0.655	1.089	1.236	-0.600	-5.905
alpha	THieF_4wk-noTrans	4	1253.058	2138.710	0.250	0.988	1.126	1.433	-0.525	0.762
alpha	THieF_8wk-4root	4	1280.800	1981.230	0.155	0.690	1.151	1.328	-0.725	-5.190

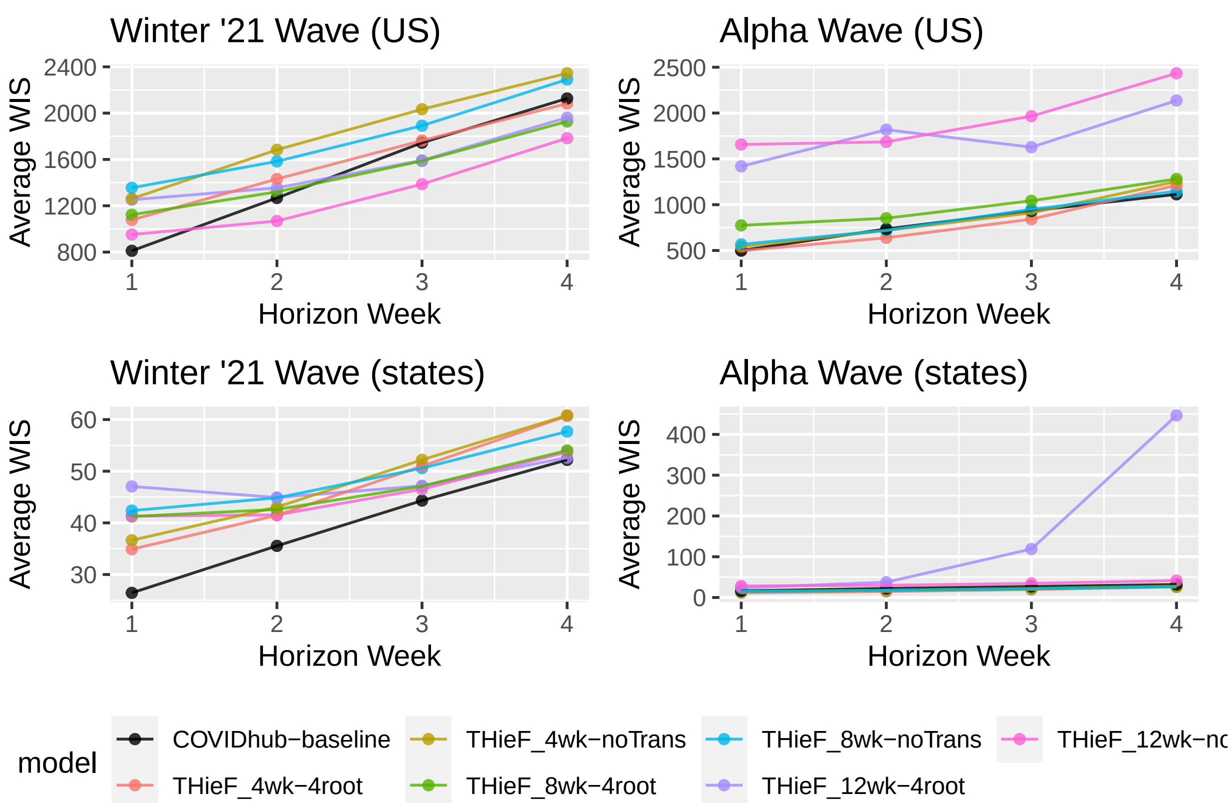
wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
alpha	THieF_12wk-4root	4	2136.839	3118.526	0.083	0.321	1.920	2.090	-0.875	-12.571
alpha	THieF_12wk-noTrans	4	2433.895	3995.532	0.071	0.786	2.187	2.677	-0.900	-3.286
winter21	COVIDhub-baseline	1	811.306	1254.871	0.686	1.000	1.000	1.000	1.000	1.000
winter21	THieF_12wk-noTrans	1	951.458	1588.678	0.175	0.905	1.173	1.266	-1.752	-0.905
winter21	THieF_4wk-4root	1	1077.810	1676.474	0.143	0.698	1.328	1.336	-1.923	-5.032
winter21	THieF_8wk-4root	1	1122.629	1789.327	0.079	0.635	1.384	1.426	-2.265	-6.302
winter21	THieF_12wk-4root	1	1251.908	1935.696	0.127	0.571	1.543	1.543	-2.009	-7.571
winter21	THieF_4wk-noTrans	1	1261.177	1824.145	0.254	0.651	1.555	1.454	-1.325	-5.984
winter21	THieF_8wk-noTrans	1	1355.462	2026.107	0.143	0.556	1.671	1.615	-1.923	-7.889
winter21	THieF_12wk-noTrans	2	1068.655	1835.087	0.190	0.921	0.843	0.899	-1.970	-0.587
winter21	COVIDhub-baseline	2	1267.664	2040.614	0.657	1.000	1.000	1.000	1.000	1.000
winter21	THieF_8wk-4root	2	1320.177	2144.319	0.127	0.810	1.041	1.051	-2.374	-2.810
winter21	THieF_12wk-4root	2	1354.690	2202.950	0.095	0.778	1.069	1.080	-2.576	-3.444
winter21	THieF_4wk-4root	2	1431.177	2051.183	0.429	0.667	1.129	1.005	-0.455	-5.667
winter21	THieF_8wk-noTrans	2	1583.384	2464.996	0.190	0.730	1.249	1.208	-1.970	-4.397
winter21	THieF_4wk-noTrans	2	1683.333	2173.562	0.492	0.651	1.328	1.065	-0.051	-5.984
winter21	THieF_12wk-noTrans	3	1386.513	2210.773	0.333	0.889	0.795	0.778	-3.889	-1.222
winter21	THieF_8wk-4root	3	1587.305	2420.965	0.302	0.746	0.910	0.852	-4.630	-4.079
winter21	THieF_12wk-4root	3	1592.375	2488.547	0.190	0.746	0.913	0.876	-7.222	-4.079
winter21	COVIDhub-baseline	3	1744.382	2840.386	0.543	1.000	1.000	1.000	1.000	1.000
winter21	THieF_4wk-4root	3	1763.142	2540.034	0.254	0.635	1.011	0.894	-5.741	-6.302
winter21	THieF_8wk-noTrans	3	1891.491	2721.611	0.429	0.667	1.084	0.958	-1.667	-5.667
winter21	THieF_4wk-noTrans	3	2033.750	2724.954	0.508	0.667	1.166	0.959	0.185	-5.667
winter21	THieF_12wk-noTrans	4	1783.649	2653.593	0.476	0.762	0.839	0.773	-1.667	-3.762
winter21	THieF_8wk-4root	4	1928.019	2853.268	0.349	0.683	0.907	0.831	-10.556	-5.349
winter21	THieF_12wk-4root	4	1961.596	2950.239	0.317	0.571	0.922	0.859	-12.778	-7.571

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
winter21	THieF_4wk-4root	4	2082.909	3023.723	0.190	0.635	0.979	0.881	-21.667	-6.302
winter21	COVIDhub-baseline	4	2126.820	3433.271	0.486	1.000	1.000	1.000	-1.000	1.000
winter21	THieF_8wk-noTrans	4	2290.411	3254.271	0.381	0.635	1.077	0.948	-8.333	-6.302
winter21	THieF_4wk-noTrans	4	2344.075	3239.480	0.381	0.667	1.102	0.944	-8.333	-5.667

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
alpha	THieF_4wk-4root	1	11.360	17.772	0.464	0.912	0.698	1.053	-0.081	-0.879
alpha	THieF_4wk-noTrans	1	13.926	22.203	0.635	0.988	0.855	1.315	0.302	0.862
alpha	THieF_8wk-noTrans	1	14.709	22.698	0.634	0.979	0.903	1.345	0.299	0.669
alpha	THieF_8wk-4root	1	15.285	23.755	0.371	0.832	0.939	1.407	-0.289	-2.695
alpha	COVIDhub-baseline	1	16.284	16.882	0.948	0.994	1.000	1.000	1.000	1.000
alpha	THieF_12wk-4root	1	23.671	33.974	0.320	0.763	1.454	2.012	-0.402	-4.276
alpha	THieF_12wk-noTrans	1	28.261	46.086	0.268	0.887	1.735	2.730	-0.518	-1.435
alpha	THieF_4wk-4root	2	14.797	23.332	0.475	0.934	0.668	1.019	-0.053	-0.369
alpha	THieF_8wk-4root	2	17.458	27.790	0.397	0.896	0.788	1.214	-0.224	-1.217
alpha	THieF_8wk-noTrans	2	17.610	26.493	0.685	0.989	0.795	1.157	0.401	0.869
alpha	THieF_4wk-noTrans	2	17.639	27.843	0.681	0.988	0.796	1.216	0.392	0.852
alpha	COVIDhub-baseline	2	22.158	22.893	0.962	0.995	1.000	1.000	1.000	1.000
alpha	THieF_12wk-noTrans	2	29.889	50.776	0.342	0.963	1.349	2.218	-0.342	0.287
alpha	THieF_12wk-4root	2	37.544	37.120	0.358	0.842	1.694	1.621	-0.308	-2.426
alpha	THieF_4wk-4root	3	19.650	30.727	0.467	0.934	0.729	1.043	-0.072	-0.369
alpha	THieF_8wk-4root	3	21.415	34.156	0.400	0.922	0.795	1.160	-0.218	-0.643
alpha	THieF_8wk-noTrans	3	21.607	33.018	0.691	0.990	0.802	1.121	0.418	0.896
alpha	THieF_4wk-noTrans	3	21.933	34.492	0.701	0.984	0.814	1.171	0.438	0.768
alpha	COVIDhub-baseline	3	26.943	29.456	0.958	0.994	1.000	1.000	1.000	1.000
alpha	THieF_12wk-noTrans	3	34.587	58.982	0.367	0.974	1.284	2.002	-0.290	0.535

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
alpha	THieF_12wk-4root	3	118.891	43.789	0.371	0.863	4.413	1.487	-0.283	-1.963
alpha	THieF_8wk-noTrans	4	26.094	40.232	0.688	0.984	0.817	1.075	0.423	0.779
alpha	THieF_4wk-4root	4	26.428	40.076	0.432	0.912	0.828	1.071	-0.153	-0.854
alpha	THieF_8wk-4root	4	26.939	42.099	0.404	0.913	0.844	1.125	-0.215	-0.833
alpha	THieF_4wk-noTrans	4	27.634	43.524	0.678	0.975	0.866	1.163	0.400	0.567
alpha	COVIDhub-baseline	4	31.920	37.413	0.945	0.994	1.000	1.000	1.000	1.000
alpha	THieF_12wk-noTrans	4	41.596	70.599	0.363	0.972	1.303	1.887	-0.308	0.496
alpha	THieF_12wk-4root	4	446.683	54.447	0.360	0.849	13.994	1.455	-0.315	-2.308
winter21	COVIDhub-baseline	1	26.445	37.213	0.783	0.977	1.000	1.000	1.000	1.000
winter21	THieF_4wk-4root	1	34.895	52.518	0.431	0.884	1.320	1.411	-0.245	-2.453
winter21	THieF_4wk-noTrans	1	36.626	52.952	0.390	0.826	1.385	1.423	-0.388	-4.600
winter21	THieF_8wk-4root	1	41.229	59.219	0.385	0.830	1.559	1.591	-0.405	-4.463
winter21	THieF_12wk-noTrans	1	41.303	57.208	0.385	0.829	1.562	1.537	-0.405	-4.504
winter21	THieF_8wk-noTrans	1	42.373	60.399	0.324	0.805	1.602	1.623	-0.622	-5.393
winter21	THieF_12wk-4root	1	47.049	66.133	0.331	0.788	1.779	1.777	-0.599	-6.015
winter21	COVIDhub-baseline	2	35.542	49.816	0.836	0.990	1.000	1.000	1.000	1.000
winter21	THieF_4wk-4root	2	41.463	63.929	0.432	0.897	1.167	1.283	-0.203	-1.308
winter21	THieF_12wk-noTrans	2	41.544	61.768	0.425	0.882	1.169	1.240	-0.223	-1.677
winter21	THieF_8wk-4root	2	42.602	64.976	0.398	0.885	1.199	1.304	-0.305	-1.600
winter21	THieF_4wk-noTrans	2	43.053	63.407	0.434	0.842	1.211	1.273	-0.197	-2.681
winter21	THieF_8wk-noTrans	2	44.848	67.870	0.345	0.855	1.262	1.362	-0.461	-2.348
winter21	THieF_12wk-4root	2	44.902	67.134	0.380	0.864	1.263	1.348	-0.356	-2.138
winter21	COVIDhub-baseline	3	44.312	64.019	0.840	0.992	1.000	1.000	1.000	1.000
winter21	THieF_12wk-noTrans	3	46.517	71.337	0.425	0.886	1.050	1.114	-0.219	-1.510
winter21	THieF_8wk-4root	3	47.090	73.430	0.405	0.890	1.063	1.147	-0.278	-1.427
winter21	THieF_12wk-4root	3	47.183	72.414	0.397	0.883	1.065	1.131	-0.302	-1.585

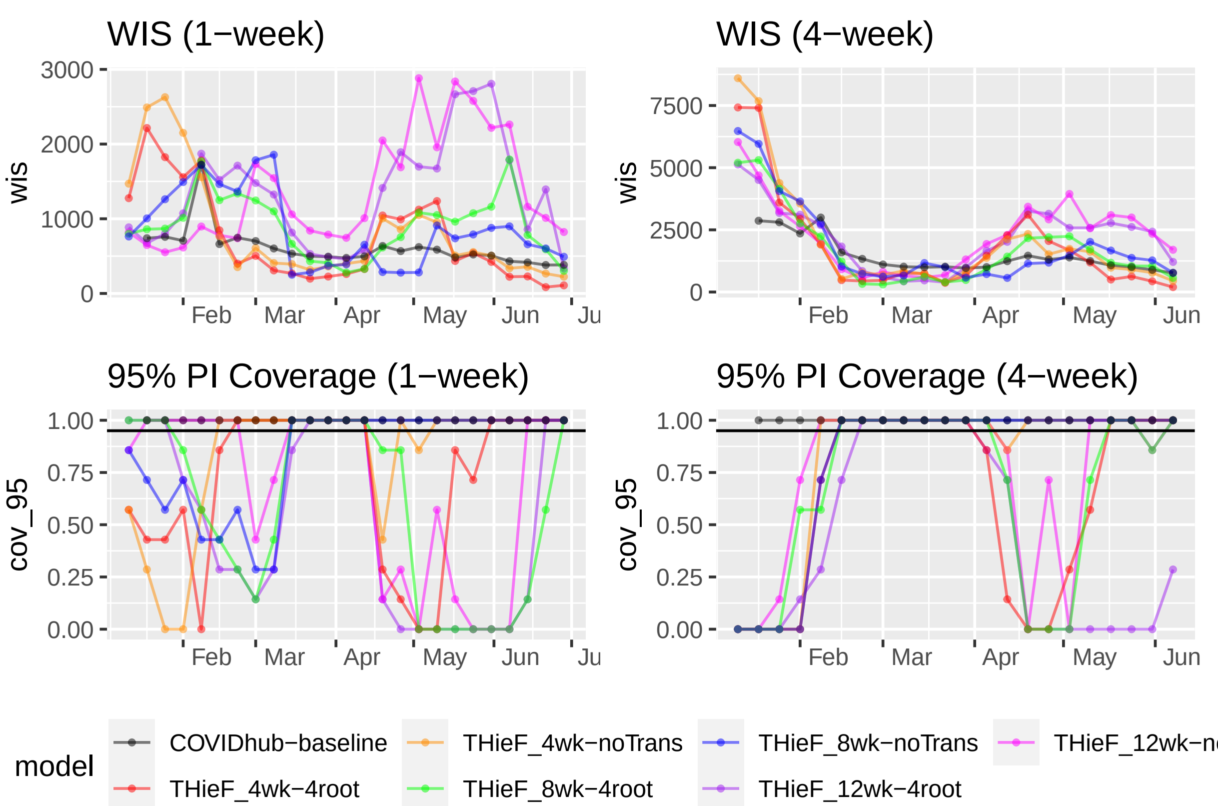
wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
winter21	THieF_8wk-noTrans	3	50.585	77.557	0.372	0.851	1.142	1.211	-0.376	-2.352
winter21	THieF_4wk-4root	3	50.967	78.386	0.431	0.881	1.150	1.224	-0.203	-1.637
winter21	THieF_4wk-noTrans	3	52.197	76.142	0.459	0.832	1.178	1.189	-0.121	-2.816
winter21	COVIDhub-baseline	4	52.196	76.859	0.839	0.990	1.000	1.000	1.000	1.000
winter21	THieF_12wk-4root	4	52.629	80.624	0.406	0.881	1.008	1.049	-0.276	-1.728
winter21	THieF_12wk-noTrans	4	53.730	82.243	0.432	0.883	1.029	1.070	-0.200	-1.691
winter21	THieF_8wk-4root	4	54.017	83.255	0.413	0.881	1.035	1.083	-0.256	-1.738
winter21	THieF_8wk-noTrans	4	57.679	87.922	0.411	0.850	1.105	1.144	-0.263	-2.505
winter21	THieF_4wk-4root	4	60.683	91.900	0.432	0.867	1.163	1.196	-0.200	-2.089
winter21	THieF_4wk-noTrans	4	60.808	88.164	0.474	0.832	1.165	1.147	-0.077	-2.968



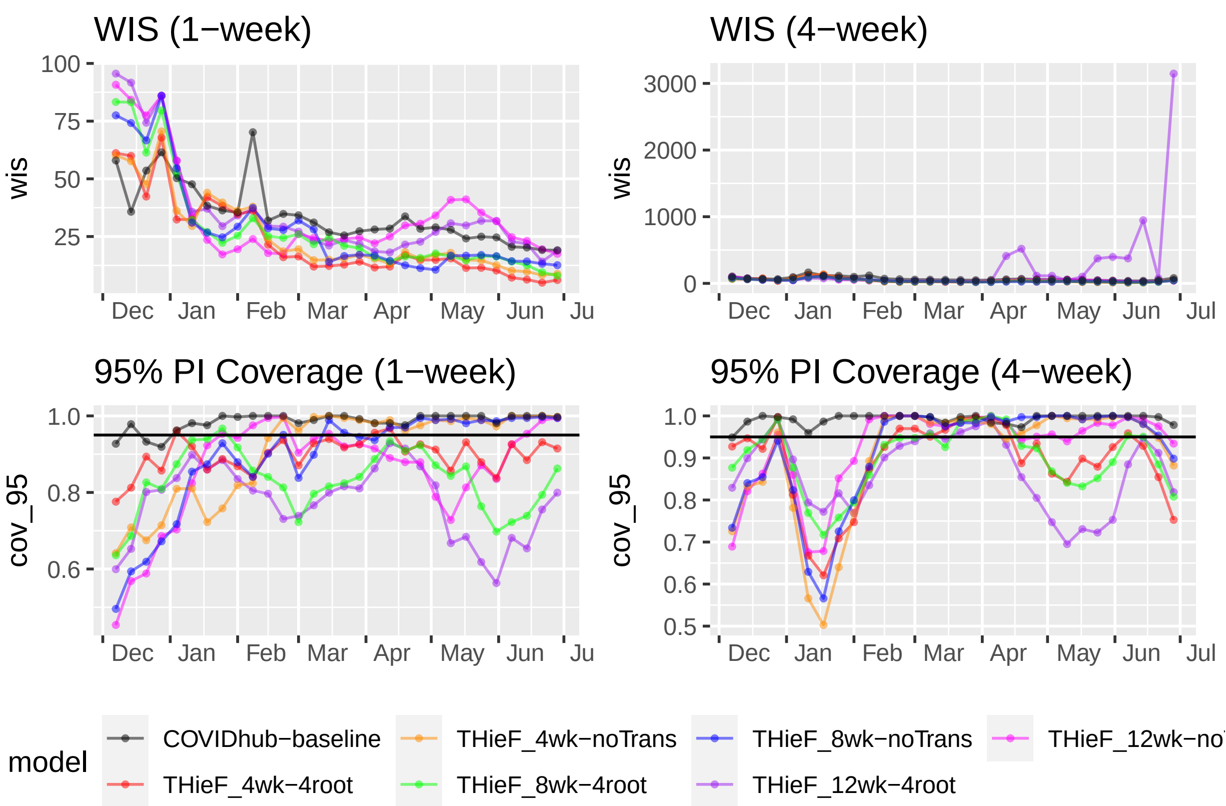
For US National during the alpha wave, the Base_arima_noTrans model performed the best for coverage rates and second best for WIS and MAE at every horizon. The Base_arima_4root model performed best in terms of WIS and MAE but was third for coverage rate while the Topmost8_arima_noTrans model had the second best coverage rates and third best WIS and MAE. During the alpha wave, these models performed the best and tended to have more consistent values for their metrics. The winter21 wave saw noticeably worse values for all metrics at every horizon. Generally, the two models that use the base set hierarchy tended to perform the best for all metrics, although the coverage rates varied a lot by horizon, with the Topmost8_arima_4root and Multiple3_arima_noTrans achieving closer to nominal coverage rates for the 95% interval than even the Base models at horizons greater than 1 week ahead. These results are pretty consistent with improvements over the base-line, which ranged from modest to significant for the best performing models during the alpha wave, especially for WIS and 50% PI coverage. The winter21 wave saw fairly comparable or worse performance compared to the baseline for WIS and MAE. The 50% coverage was better than the baseline for all models, the greatest improvements for the best models, but the 95% coverage could be better or worse for the best models.

For states during both waves, we again see noticeably better coverage rates that approach the nominal level (or even exceed it) for all of the models, compared to those of the US National level when broken into waves. During the alpha wave, the Base_arima_4root model has the best performance for all metrics across all horizons. It is followed closely by the Topmost8_arima_noTrans and Base_arima_noTrans models, then the Topmost8_arima_4root and Multiple3_arima_noTrans models, all of which are much better than the Multiple3_arima_4root model. During the winter21 wave, the Base_arima_4root model is again generally the best for all metrics at all horizon weeks but sometimes loses out in terms of 95% coverage at longer horizons to other models. Notably during this wave, there are much greater similarities in coverage rates and certain models have better coverage rates for the 50% interval while others have better coverage reads for the 95% one. Similar patterns to the ones described above for the US national level when compared to the baseline were observed.

By forecast date (week)



```
## 'summarise()' has grouped output by 'forecast_date', 'model'. You can override
## using the '.groups' argument.
## 'summarise()' has grouped output by 'forecast_date', 'model'. You can override
## using the '.groups' argument.
```



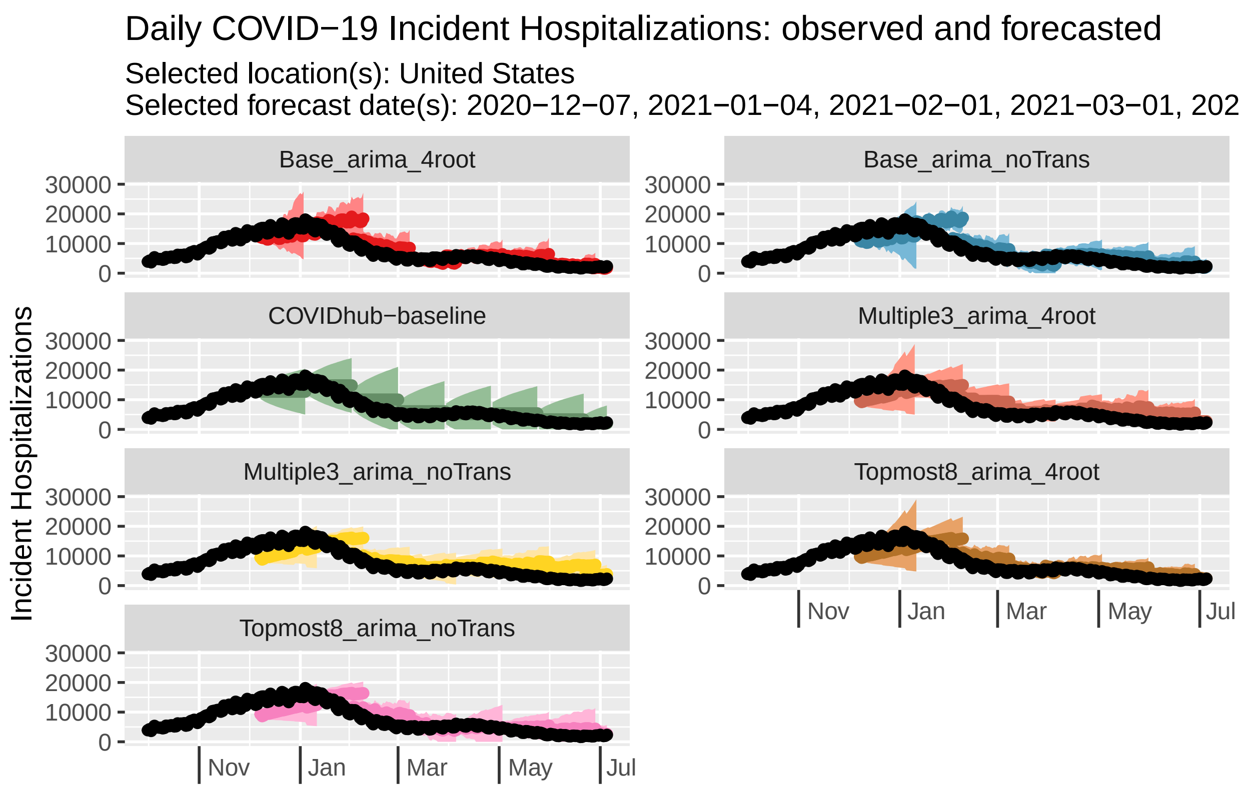
Scoring Conclusions

It seems that the best three models that should be passed onto the testing phase are the Base_arima_noTrans, the Base_arima_4root, and the Topmost8_arima_noTrans models, which seem to take turns being most accurate depending on state or national level overall and during different waves.

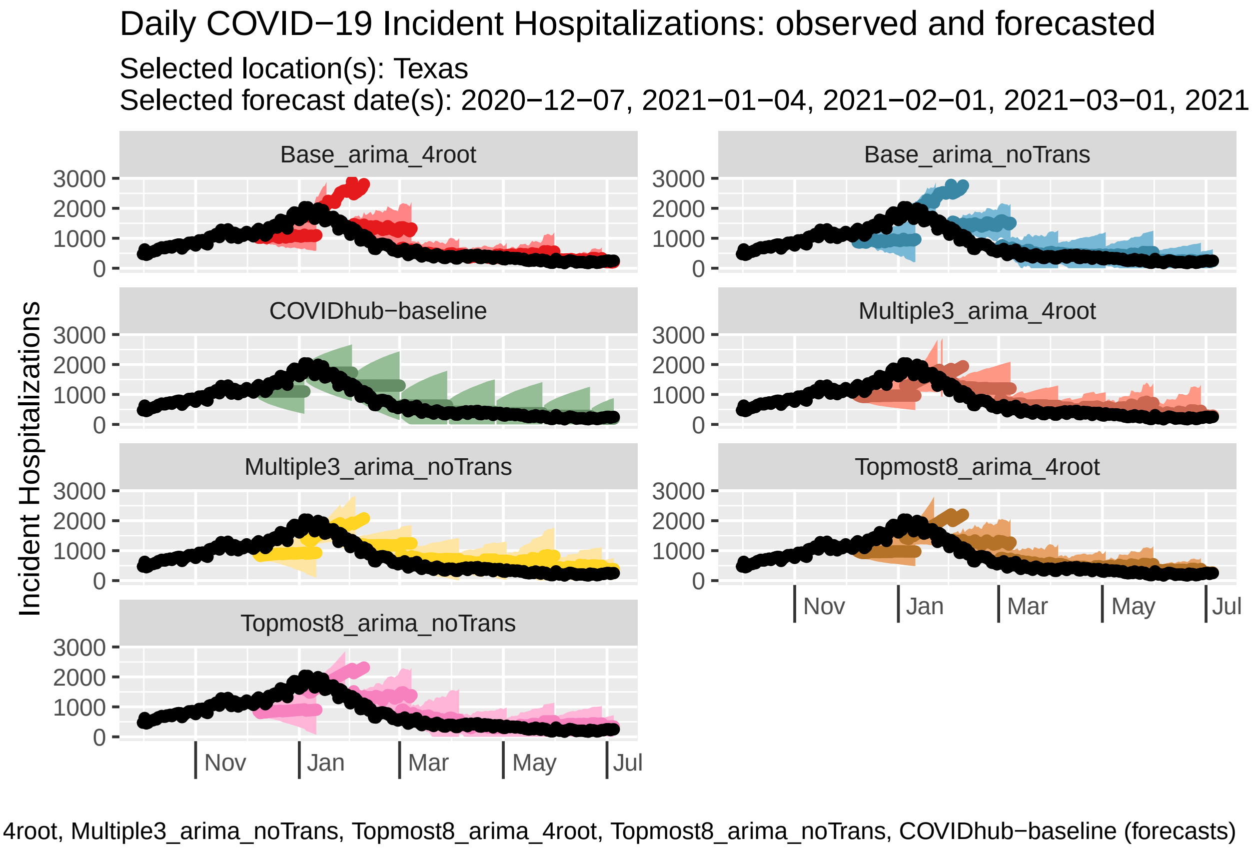
Original Results

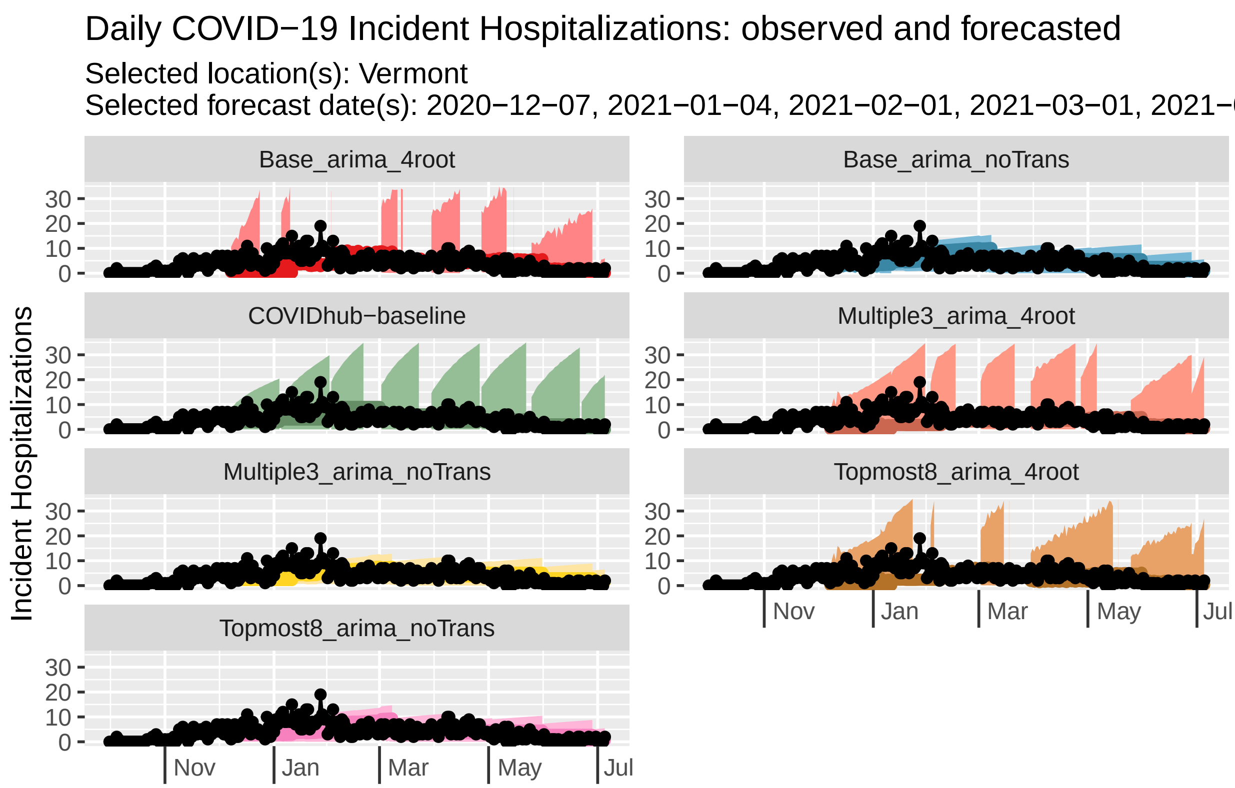
Plot Forecasts

We plot forecasts for three locations (Us National, Texas, and Vermont) for each of the three models. The two state locations were chosen as the highest and lowest hospitalization count locations. The plotted forecasts are spaced 4 weeks apart with 1 - 28 day ahead horizons, starting with the first forecast date, 12/07/20, until the end of the training period, 07/05/21.



4root, Multiple3_arima_noTrans, Topmost8_arima_4root, Topmost8_arima_noTrans, COVIDhub−baseline (forecasts)


Daily COVID−19 Incident Hospitalizations: observed and forecasted
Selected location(s): Texas
Selected forecast date(s): 2020−12−07, 2021−01−04, 2021−02−01, 2021−03−01, 2021
Base_arima_4root
Base_arima_noTrans
COVIDhub−baseline
Multiple3_arima_4root
Multiple3_arima_noTrans
Topmost8_arima_4root
Topmost8_arima_noTrans
Incident Hospitalizations
3000
2000
1000
0
Nov
Jan
Mar
May
Jul
4root, Multiple3_arima_noTrans, Topmost8_arima_4root, Topmost8_arima_noTrans, COVIDhub−baseline (forecasts)



4root, Multiple3_arima_noTrans, Topmost8_arima_4root, Topmost8_arima_noTrans, COVIDhub−baseline (forecasts)

The plots don't show much difference between the 6 models (although this is partially due to the fact that the yscales are free and are not consistent even for the same location). However, of note is the fact that the models that use a forth root transformation tend to have wider prediction intervals than the models without transformations. All of the models reasonably follow the trend of the truth lines for every location shown, although it seems like lower count locations may have slightly greater accuracy. The baseline seems to have intervals of fairly consistent widths while the models tend to vary in their interval widths, tending to show wider intervals during times of higher incidents or change points and narrow lower intervals otherwise. We investigate model accuracy more deeply by scoring the forecasts for each model and calculating appropriate metrics.

Score forecasts

We calculate metrics for all three models, aggregating by horizon. Since hospitalizations are forecasted at a daily time scale, a new horizon week variable is created in which scoring metrics are averaged over from a single week Tuesday to the following Monday. We perform these calculations for the entire training phase as well as for each of the two waves that occurred during the training phase.

Overall metrics

model	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
Base_arima_4root	1029.555	1564.399	0.316	0.816	0.915	0.939	-0.710	-2.826
Base_arima_noTrans	1095.971	1597.582	0.401	0.838	0.974	0.959	-0.382	-2.351
COVIDhub-baseline	1125.784	1665.169	0.759	0.997	1.000	1.000	1.000	1.000
Topmost8_arima_4root	1193.043	1835.486	0.207	0.668	1.060	1.102	-1.130	-5.939

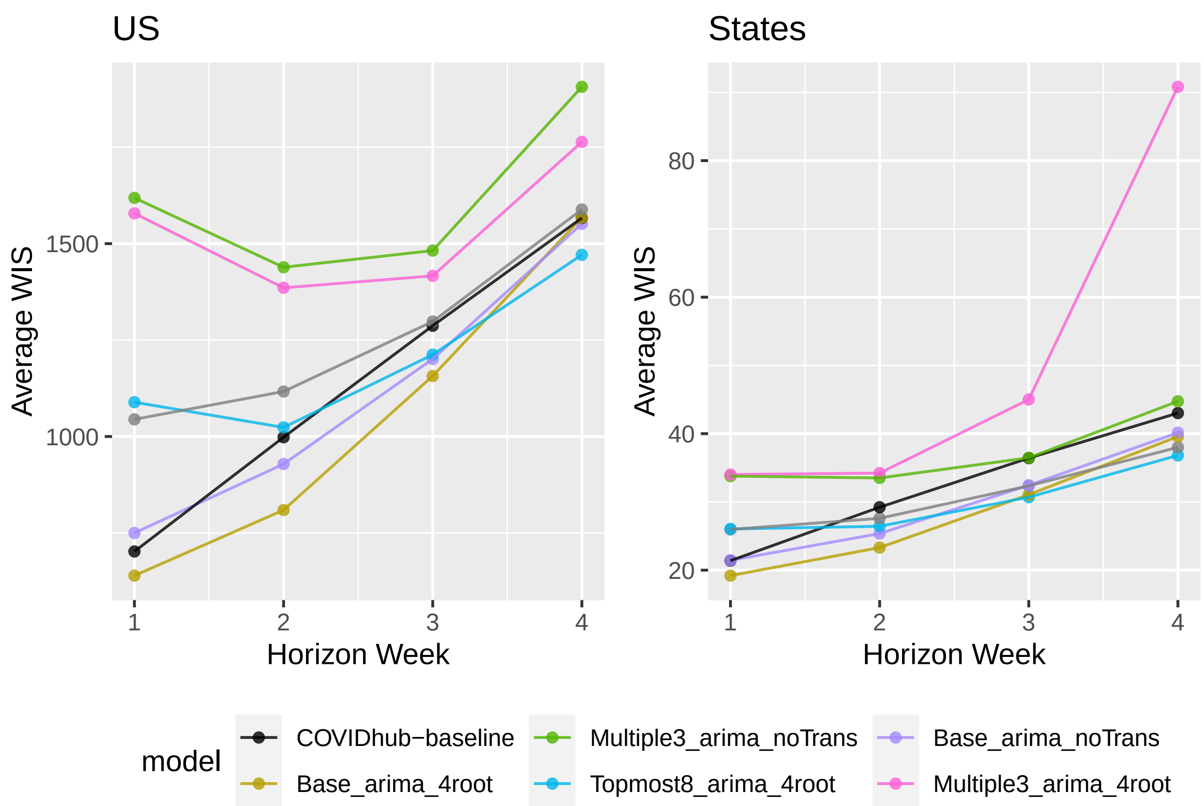
model	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
Topmost8_arima_noTrans	1253.977	1926.286	0.305	0.806	1.114	1.157	-0.754	-3.037
Multiple3_arima_4root	1533.491	2274.958	0.164	0.564	1.362	1.366	-1.295	-8.129
Multiple3_arima_noTrans	1607.525	2471.404	0.154	0.702	1.428	1.484	-1.333	-5.227

model	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
Base_arima_4root	27.986	42.612	0.481	0.916	0.869	1.011	-0.050	-0.850
Base_arima_noTrans	29.573	43.962	0.568	0.916	0.919	1.043	0.175	-0.846
Topmost8_arima_4root	29.835	45.381	0.413	0.882	0.927	1.076	-0.224	-1.680
Topmost8_arima_noTrans	30.801	46.088	0.522	0.913	0.957	1.093	0.058	-0.904
COVIDhub-baseline	32.196	42.158	0.887	0.990	1.000	1.000	1.000	1.000
Multiple3_arima_noTrans	36.968	57.564	0.376	0.904	1.148	1.365	-0.321	-1.151
Multiple3_arima_4root	50.220	52.952	0.364	0.836	1.560	1.256	-0.352	-2.833

Horizon week metrics

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
Base_arima_4root	1	639.347	972.010	0.257	0.786	0.911	0.994	-0.895	-4.059
COVIDhub-baseline	1	701.913	978.105	0.771	0.990	1.000	1.000	1.000	1.000
Base_arima_noTrans	1	749.818	1114.862	0.295	0.824	1.068	1.140	-0.754	-3.118
Base_arima_4root	2	809.315	1255.389	0.409	0.877	0.811	0.866	-0.311	-1.463
Base_arima_noTrans	2	928.603	1332.619	0.453	0.828	0.930	0.920	-0.160	-2.448
COVIDhub-baseline	2	998.101	1448.882	0.793	1.000	1.000	1.000	1.000	1.000
Topmost8_arima_4root	2	1023.521	1616.694	0.217	0.778	1.025	1.116	-0.966	-3.433
Topmost8_arima_noTrans	1	1044.468	1506.746	0.219	0.733	1.488	1.540	-1.035	-5.353
Topmost8_arima_4root	1	1088.981	1550.934	0.114	0.467	1.551	1.586	-1.421	-11.941
Topmost8_arima_noTrans	2	1116.592	1740.354	0.305	0.818	1.119	1.201	-0.664	-2.645
Base_arima_4root	3	1156.882	1765.098	0.337	0.827	0.899	0.914	-0.640	-2.469
Base_arima_noTrans	3	1200.986	1718.652	0.490	0.847	0.933	0.890	-0.040	-2.061
Topmost8_arima_4root	3	1212.164	1919.171	0.250	0.755	0.942	0.994	-0.980	-3.898
COVIDhub-baseline	3	1287.121	1930.429	0.755	1.000	1.000	1.000	1.000	1.000
Topmost8_arima_noTrans	3	1298.125	2080.367	0.321	0.847	1.009	1.078	-0.700	-2.061
Multiple3_arima_4root	2	1385.549	2098.428	0.158	0.626	1.388	1.448	-1.168	-6.488
Multiple3_arima_4root	3	1416.465	2197.313	0.224	0.709	1.100	1.138	-1.080	-4.816
Multiple3_arima_noTrans	2	1438.803	2266.634	0.108	0.739	1.442	1.564	-1.336	-4.222
Topmost8_arima_4root	4	1470.919	2299.869	0.254	0.683	0.939	0.964	-1.148	-5.349
Multiple3_arima_noTrans	3	1482.037	2403.128	0.219	0.852	1.151	1.245	-1.100	-1.959
Base_arima_noTrans	4	1551.446	2292.973	0.370	0.857	0.990	0.961	-0.605	-1.857
COVIDhub-baseline	4	1566.580	2385.799	0.714	1.000	1.000	1.000	1.000	1.000
Base_arima_4root	4	1567.629	2346.378	0.259	0.772	1.001	0.983	-1.123	-3.550
Multiple3_arima_4root	1	1578.437	2140.209	0.052	0.352	2.249	2.188	-1.649	-14.765
Topmost8_arima_noTrans	4	1588.545	2432.356	0.381	0.831	1.014	1.020	-0.556	-2.386
Multiple3_arima_noTrans	1	1618.621	2246.387	0.095	0.471	2.306	2.297	-1.491	-11.824
Multiple3_arima_4root	4	1763.813	2694.806	0.233	0.582	1.126	1.130	-1.247	-7.360
Multiple3_arima_noTrans	4	1906.551	3012.165	0.201	0.762	1.217	1.263	-1.395	-3.762

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
Base_arima_4root	1	19.203	28.513	0.478	0.901	0.898	1.053	-0.060	-1.403
COVIDhub-baseline	1	21.392	27.090	0.865	0.985	1.000	1.000	1.000	1.000
Base_arima_noTrans	1	21.445	31.406	0.522	0.911	1.002	1.159	0.060	-1.107
Base_arima_4root	2	23.308	35.774	0.502	0.932	0.798	0.968	0.004	-0.415
Base_arima_noTrans	2	25.362	38.117	0.580	0.928	0.868	1.031	0.203	-0.528
Topmost8_arima_noTrans	1	25.969	36.814	0.478	0.885	1.214	1.359	-0.060	-1.869
Topmost8_arima_4root	1	26.038	37.306	0.379	0.813	1.217	1.377	-0.332	-3.921
Topmost8_arima_4root	2	26.430	40.325	0.427	0.899	0.904	1.091	-0.184	-1.216
Topmost8_arima_noTrans	2	27.599	40.954	0.535	0.924	0.944	1.108	0.087	-0.604
COVIDhub-baseline	2	29.225	36.963	0.896	0.992	1.000	1.000	1.000	1.000
Topmost8_arima_4root	3	30.690	47.891	0.425	0.914	0.843	0.999	-0.191	-0.830
Base_arima_4root	3	31.057	47.771	0.480	0.923	0.853	0.997	-0.051	-0.622
Topmost8_arima_noTrans	3	32.357	49.562	0.532	0.928	0.889	1.034	0.082	-0.521
Base_arima_noTrans	3	32.443	48.423	0.590	0.918	0.891	1.010	0.227	-0.741
Multiple3_arima_noTrans	2	33.509	52.085	0.380	0.922	1.147	1.409	-0.303	-0.669
Multiple3_arima_noTrans	1	33.784	48.465	0.326	0.827	1.579	1.789	-0.477	-3.518
Multiple3_arima_4root	1	34.017	46.974	0.318	0.745	1.590	1.734	-0.500	-5.869
Multiple3_arima_4root	2	34.219	48.861	0.369	0.847	1.171	1.322	-0.330	-2.437
COVIDhub-baseline	3	36.412	47.923	0.896	0.993	1.000	1.000	1.000	1.000
Multiple3_arima_noTrans	3	36.466	58.694	0.408	0.940	1.002	1.225	-0.231	-0.237
Topmost8_arima_4root	4	36.822	57.179	0.425	0.908	0.856	0.977	-0.191	-1.005
Topmost8_arima_noTrans	4	37.998	58.305	0.548	0.919	0.883	0.997	0.124	-0.741
Base_arima_4root	4	39.583	60.273	0.462	0.906	0.920	1.030	-0.096	-1.049
Base_arima_noTrans	4	40.149	59.564	0.581	0.906	0.933	1.018	0.207	-1.059
COVIDhub-baseline	4	43.018	58.501	0.891	0.992	1.000	1.000	1.000	1.000
Multiple3_arima_noTrans	4	44.741	72.387	0.394	0.931	1.040	1.237	-0.272	-0.445
Multiple3_arima_4root	3	45.006	52.951	0.392	0.884	1.236	1.105	-0.272	-1.533
Multiple3_arima_4root	4	90.817	63.989	0.380	0.873	2.111	1.094	-0.306	-1.825



When splitting the results so that US National is separate, we find that for US National the two models that use the base set hierarchy tended to perform the best, with improvements seen over a naive baseline in all metrics except for 95%PI Coverage. These two models are followed by the Topmost8_arima_noTrans model, which showed more comparable performance to the baseline. The Base_arima_noTrans model performed the best in terms of PI Coverage, followed by the Base_arima_4root and Topmost8_arima_noTrans models (with the latter being more consistent with coverage rates for all horizons). In terms of WIS and MAE, the Base_arima_4root model performed the best, followed by the Base_arima_noTrans model. Overall, for the US National level, it seems that there is a pattern that for coverage rates, the models without transformations perform better (and these tend to have smaller prediction intervals) while for WIS and MAE the fourth root transform models perform better; these patterns hold at all horizons. Further, there is a general order to model accuracy in terms of hierarchical structure used: the Base set performs the best, followed by the Topmost8 set, and finally the Multiple3 set.

For the state level, we see noticeably better coverage rates that approach the nominal level (or even exceed it) for all of the models. (We also see shrunken WIS and MAE values, although this is likely partially due to the fact that the states have smaller scale values than the National level.) Interestingly, we see fewer improvements to MAE over the baseline but more substantial improvements in terms of coverage rates. It's a little harder to definitively say which models have the best coverage rates, as the results differ between the 50% and 95% intervals and the models are quite similar in coverage rates, especially for the 95% intervals. However, it seems that the Base_arima_4root model has the best performance for all metrics across all horizons. It is followed closely by the Topmost8_arima_noTrans and Base_arima_noTrans models, then the Topmost8_arima_4root and Multiple3_arima_noTrans models, all of which are much better than the Multiple3_arima_4root model.

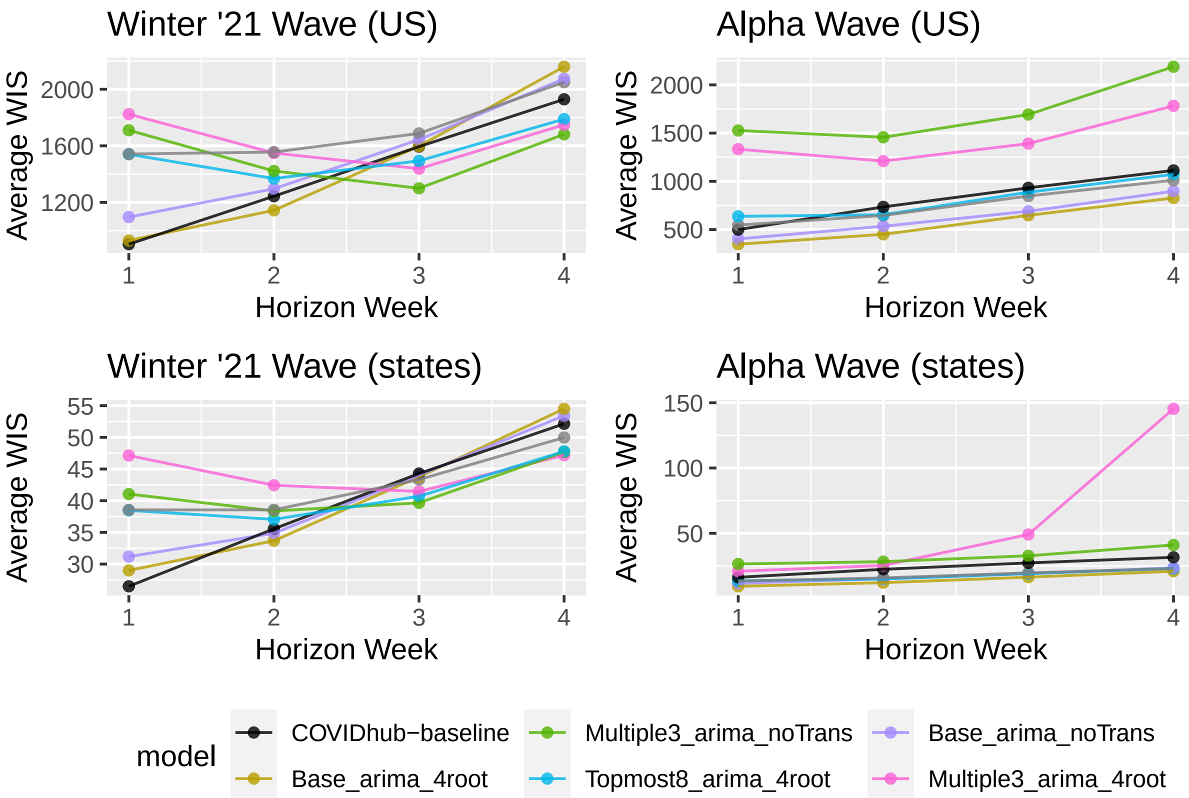
Metrics by wave

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
alpha	Base_arima_4root	1	348.229	543.021	0.295	0.819	0.697	1.019	-0.434	-2.619
alpha	Base_arima_noTrans	1	403.427	678.426	0.419	1.000	0.807	1.274	-0.172	1.000
alpha	Base_arima_4root	2	450.928	703.528	0.429	0.939	0.613	0.833	-0.146	-0.224
alpha	COVIDhub-baseline	1	499.666	532.705	0.971	1.000	1.000	1.000	1.000	1.000
alpha	Base_arima_noTrans	2	534.593	827.925	0.510	1.000	0.727	0.980	0.021	1.000
alpha	Topmost8_arima_noTrans	1	546.701	952.788	0.238	0.981	1.094	1.789	-0.556	0.619
alpha	Topmost8_arima_4root	1	637.366	943.945	0.114	0.467	1.276	1.772	-0.818	-9.667
alpha	Topmost8_arima_noTrans	2	645.899	1124.302	0.439	1.000	0.879	1.331	-0.125	1.000
alpha	Base_arima_4root	3	647.750	1044.362	0.319	0.901	0.695	0.893	-0.371	-0.978
alpha	Topmost8_arima_4root	2	653.822	1010.887	0.265	0.735	0.890	1.196	-0.479	-4.306
alpha	Base_arima_noTrans	3	689.803	1088.364	0.549	1.000	0.740	0.931	0.101	1.000
alpha	COVIDhub-baseline	2	735.043	844.888	0.990	1.000	1.000	1.000	1.000	1.000
alpha	Base_arima_4root	4	827.597	1349.386	0.226	0.857	0.743	0.904	-0.575	-1.857
alpha	Topmost8_arima_noTrans	3	848.269	1468.658	0.484	1.000	0.910	1.256	-0.034	1.000
alpha	Topmost8_arima_4root	3	887.183	1363.911	0.220	0.670	0.951	1.166	-0.573	-5.593
alpha	Base_arima_noTrans	4	897.845	1508.112	0.440	1.000	0.807	1.011	-0.125	1.000
winter21	COVIDhub-baseline	1	904.160	1423.505	0.571	0.981	1.000	1.000	1.000	1.000
winter21	Base_arima_4root	1	930.466	1401.000	0.219	0.752	1.029	0.984	-3.933	-6.385
alpha	COVIDhub-baseline	3	932.443	1169.429	0.989	1.000	1.000	1.000	1.000	1.000
alpha	Topmost8_arima_noTrans	4	1011.064	1649.833	0.476	1.000	0.908	1.106	-0.050	1.000
alpha	Topmost8_arima_4root	4	1072.725	1672.564	0.190	0.619	0.964	1.121	-0.650	-6.619
winter21	Base_arima_noTrans	1	1096.208	1551.297	0.171	0.648	1.212	1.090	-4.600	-9.769
alpha	COVIDhub-baseline	4	1113.129	1492.357	0.976	1.000	1.000	1.000	1.000	1.000
winter21	Base_arima_4root	2	1143.809	1770.459	0.390	0.819	0.920	0.880	-1.000	-2.619
alpha	Multiple3_arima_4root	2	1209.980	1749.967	0.173	0.551	1.646	2.071	-0.667	-7.980
winter21	COVIDhub-baseline	2	1243.621	2012.610	0.610	1.000	1.000	1.000	1.000	1.000
winter21	Base_arima_noTrans	2	1296.345	1803.667	0.400	0.667	1.042	0.896	-0.913	-5.667
winter21	Multiple3_arima_noTrans	3	1299.539	2099.034	0.276	0.819	0.815	0.810	-4.273	-2.619
alpha	Multiple3_arima_4root	1	1333.092	1769.323	0.029	0.324	2.668	3.321	-1.000	-12.524
winter21	Topmost8_arima_4root	2	1368.573	2182.115	0.171	0.819	1.100	1.084	-3.000	-2.619
alpha	Multiple3_arima_4root	3	1391.170	2040.682	0.154	0.495	1.492	1.745	-0.708	-9.110
winter21	Multiple3_arima_noTrans	2	1422.074	2181.710	0.133	0.771	1.143	1.084	-3.348	-3.571
winter21	Multiple3_arima_4root	3	1438.387	2333.060	0.286	0.895	0.902	0.901	-4.091	-1.095
alpha	Multiple3_arima_noTrans	2	1456.727	2357.624	0.082	0.704	1.982	2.790	-0.854	-4.918
winter21	Topmost8_arima_4root	3	1493.815	2400.396	0.276	0.829	0.937	0.927	-4.273	-2.429
alpha	Multiple3_arima_noTrans	1	1526.898	2190.979	0.086	0.448	3.056	4.113	-0.879	-10.048
winter21	Topmost8_arima_4root	1	1540.596	2157.924	0.114	0.467	1.704	1.516	-5.400	-15.615
winter21	Topmost8_arima_noTrans	1	1542.234	2060.705	0.200	0.486	1.706	1.448	-4.200	-15.000
winter21	Multiple3_arima_4root	2	1549.413	2423.658	0.143	0.695	1.246	1.204	-3.261	-5.095
winter21	Topmost8_arima_noTrans	2	1555.905	2315.335	0.181	0.648	1.251	1.150	-2.913	-6.048
winter21	COVIDhub-baseline	3	1594.509	2589.962	0.552	1.000	1.000	1.000	1.000	1.000

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
winter21	Base_arima_4root	3	1598.130	2389.736	0.352	0.762	1.002	0.923	-2.818	-3.762
winter21	Base_arima_noTrans	3	1644.012	2264.901	0.438	0.714	1.031	0.874	-1.182	-4.714
winter21	Multiple3_arima_noTrans	4	1681.499	2593.573	0.314	0.781	0.872	0.836	-39.000	-3.381
winter21	Topmost8_arima_noTrans	3	1688.000	2610.515	0.181	0.714	1.059	1.008	-6.091	-4.714
alpha	Multiple3_arima_noTrans	3	1692.610	2754.007	0.154	0.890	1.815	2.355	-0.708	-1.198
winter21	Multiple3_arima_noTrans	1	1710.345	2301.795	0.105	0.495	1.892	1.617	-5.533	-14.692
winter21	Multiple3_arima_4root	4	1748.996	2739.448	0.324	0.762	0.907	0.884	-37.000	-3.762
alpha	Multiple3_arima_4root	4	1782.333	2639.003	0.119	0.357	1.601	1.768	-0.800	-11.857
winter21	Topmost8_arima_4root	4	1789.474	2801.714	0.305	0.733	0.928	0.904	-41.000	-4.333
winter21	Multiple3_arima_4root	1	1823.782	2511.095	0.076	0.381	2.017	1.764	-5.933	-18.385
winter21	COVIDhub-baseline	4	1929.341	3100.552	0.505	1.000	1.000	1.000	1.000	1.000
winter21	Topmost8_arima_noTrans	4	2050.529	3058.375	0.305	0.695	1.063	0.986	-41.000	-5.095
winter21	Base_arima_noTrans	4	2074.327	2920.862	0.314	0.743	1.075	0.942	-39.000	-4.143
winter21	Base_arima_4root	4	2159.655	3143.972	0.286	0.705	1.119	1.014	-45.000	-4.905
alpha	Multiple3_arima_noTrans	4	2187.866	3535.404	0.060	0.738	1.966	2.369	-0.925	-4.238

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
alpha	Base_arima_4root	1	9.401	14.533	0.497	0.916	0.577	0.861	-0.007	-0.778
alpha	Base_arima_noTrans	1	11.705	18.225	0.657	0.990	0.719	1.080	0.349	0.904
alpha	Base_arima_4root	2	12.182	18.808	0.520	0.950	0.543	0.814	0.043	0.008
alpha	Topmost8_arima_noTrans	1	13.384	20.877	0.608	0.974	0.822	1.237	0.241	0.544
alpha	Topmost8_arima_4root	1	13.596	20.846	0.382	0.819	0.835	1.235	-0.264	-3.000
alpha	Topmost8_arima_4root	2	15.042	23.943	0.423	0.905	0.671	1.036	-0.167	-1.006
alpha	Base_arima_noTrans	2	15.171	23.447	0.712	0.993	0.676	1.014	0.458	0.965
alpha	Topmost8_arima_noTrans	2	15.852	23.423	0.702	0.988	0.707	1.013	0.438	0.849
alpha	COVIDhub-baseline	1	16.284	16.882	0.948	0.994	1.000	1.000	1.000	1.000
alpha	Base_arima_4root	3	16.394	25.759	0.502	0.955	0.600	0.877	0.005	0.109
alpha	Base_arima_noTrans	3	19.082	29.607	0.728	0.990	0.699	1.009	0.494	0.918
alpha	Topmost8_arima_4root	3	19.134	30.761	0.411	0.929	0.700	1.048	-0.193	-0.491
alpha	Topmost8_arima_noTrans	3	19.691	29.793	0.704	0.992	0.721	1.015	0.442	0.956
alpha	Multiple3_arima_4root	1	20.895	30.369	0.311	0.735	1.283	1.799	-0.421	-4.908
alpha	Base_arima_4root	4	20.933	32.675	0.490	0.950	0.662	0.917	-0.023	0.002
alpha	COVIDhub-baseline	2	22.429	23.117	0.962	0.994	1.000	1.000	1.000	1.000
alpha	Topmost8_arima_noTrans	4	23.027	35.651	0.716	0.996	0.728	1.000	0.475	1.026
alpha	Topmost8_arima_4root	4	23.110	37.052	0.407	0.936	0.731	1.039	-0.204	-0.317
alpha	Base_arima_noTrans	4	23.545	36.957	0.715	0.989	0.745	1.037	0.472	0.877
alpha	Multiple3_arima_4root	2	25.404	34.263	0.356	0.839	1.133	1.482	-0.311	-2.499

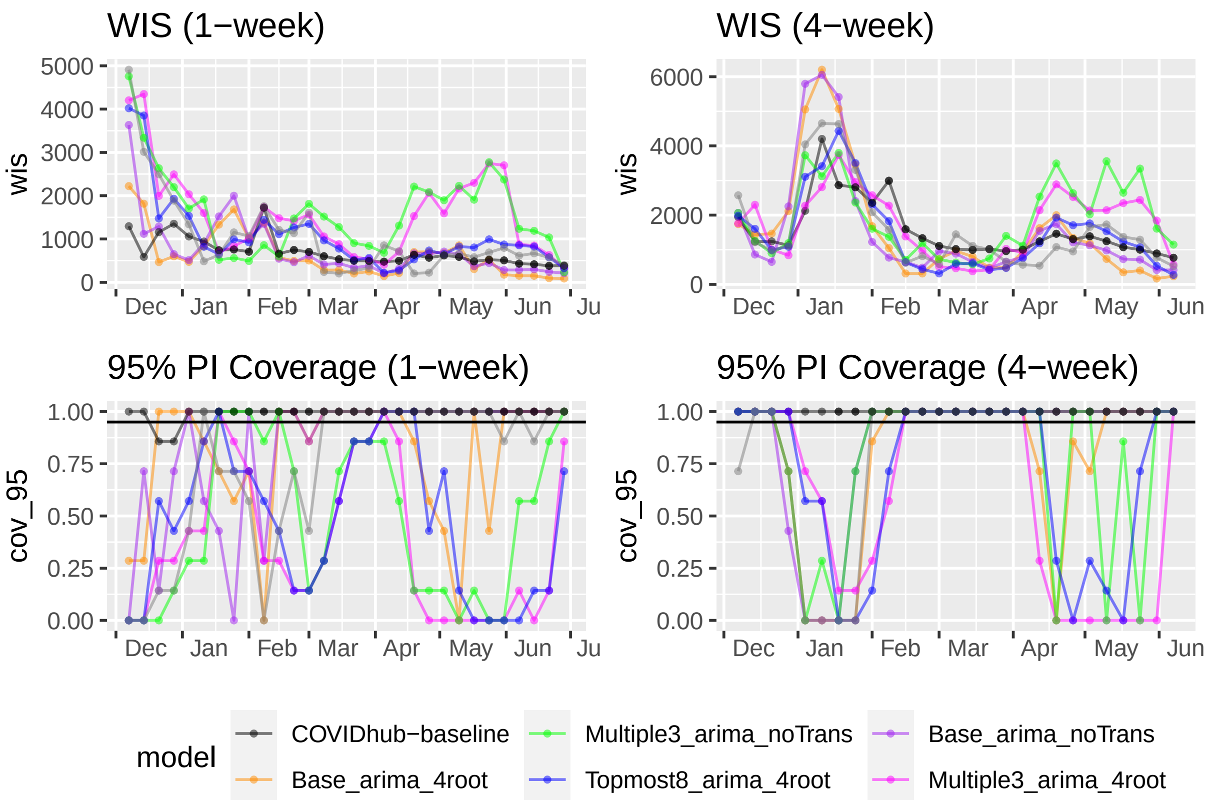
wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
winter21	COVIDhub-baseline	1	26.499	37.298	0.781	0.976	1.000	1.000	1.000	1.000
alpha	Multiple3_arima_noTrans	1	26.512	41.941	0.297	0.858	1.628	2.484	-0.453	-2.096
alpha	COVIDhub-baseline	3	27.315	29.355	0.961	0.994	1.000	1.000	1.000	1.000
alpha	Multiple3_arima_noTrans	2	28.285	47.973	0.330	0.964	1.261	2.075	-0.367	0.318
winter21	Base_arima_4root	1	29.005	42.493	0.459	0.886	1.095	1.139	-0.145	-2.448
winter21	Base_arima_noTrans	1	31.184	44.588	0.387	0.833	1.177	1.195	-0.401	-4.469
alpha	COVIDhub-baseline	4	31.616	35.644	0.956	0.995	1.000	1.000	1.000	1.000
alpha	Multiple3_arima_noTrans	3	32.778	56.299	0.356	0.977	1.200	1.918	-0.312	0.622
winter21	Base_arima_4root	2	33.692	51.608	0.485	0.916	0.947	1.035	-0.045	-0.850
winter21	Base_arima_noTrans	2	34.874	51.808	0.458	0.867	0.981	1.039	-0.125	-2.059
winter21	COVIDhub-baseline	2	35.568	49.886	0.835	0.990	1.000	1.000	1.000	1.000
winter21	Topmost8_arima_4root	2	37.059	55.615	0.431	0.892	1.042	1.115	-0.205	-1.432
winter21	Multiple3_arima_noTrans	2	38.385	55.923	0.426	0.882	1.079	1.121	-0.220	-1.682
winter21	Topmost8_arima_4root	1	38.481	53.766	0.376	0.807	1.452	1.442	-0.442	-5.462
winter21	Topmost8_arima_noTrans	1	38.553	52.752	0.348	0.795	1.455	1.414	-0.541	-5.902
winter21	Topmost8_arima_noTrans	2	38.564	57.316	0.378	0.866	1.084	1.149	-0.364	-2.095
winter21	Multiple3_arima_noTrans	3	39.663	60.769	0.454	0.908	0.895	0.949	-0.136	-1.009
winter21	Topmost8_arima_4root	3	40.706	62.737	0.436	0.902	0.919	0.980	-0.188	-1.135
winter21	Multiple3_arima_noTrans	1	41.056	54.989	0.355	0.796	1.549	1.474	-0.515	-5.895
alpha	Multiple3_arima_noTrans	4	41.065	70.591	0.333	0.977	1.299	1.980	-0.366	0.609
winter21	Multiple3_arima_4root	3	41.464	63.327	0.420	0.898	0.936	0.989	-0.236	-1.230
winter21	Multiple3_arima_4root	2	42.446	62.486	0.381	0.854	1.193	1.253	-0.355	-2.373
winter21	Topmost8_arima_noTrans	3	43.334	66.696	0.384	0.872	0.978	1.042	-0.342	-1.848
winter21	Base_arima_4root	3	43.765	66.848	0.460	0.896	0.988	1.044	-0.117	-1.278
winter21	Base_arima_noTrans	3	44.023	64.730	0.470	0.856	0.994	1.011	-0.088	-2.230
winter21	COVIDhub-baseline	3	44.295	64.016	0.839	0.992	1.000	1.000	1.000	1.000
winter21	Multiple3_arima_4root	1	47.138	63.579	0.324	0.754	1.779	1.705	-0.625	-7.476
winter21	Multiple3_arima_4root	4	47.184	72.727	0.418	0.887	0.905	0.947	-0.241	-1.583
winter21	Multiple3_arima_noTrans	4	47.682	73.824	0.442	0.895	0.914	0.961	-0.170	-1.385
winter21	Topmost8_arima_4root	4	47.791	73.280	0.440	0.885	0.917	0.954	-0.177	-1.619
alpha	Multiple3_arima_4root	3	49.094	40.979	0.361	0.868	1.797	1.396	-0.302	-1.870
winter21	Topmost8_arima_noTrans	4	49.975	76.428	0.414	0.858	0.958	0.995	-0.254	-2.317
winter21	COVIDhub-baseline	4	52.140	76.787	0.839	0.990	1.000	1.000	1.000	1.000
winter21	Base_arima_noTrans	4	53.433	77.649	0.474	0.839	1.025	1.011	-0.077	-2.784
winter21	Base_arima_4root	4	54.503	82.352	0.441	0.871	1.045	1.072	-0.175	-1.986
alpha	Multiple3_arima_4root	4	145.358	53.066	0.333	0.857	4.598	1.489	-0.367	-2.097

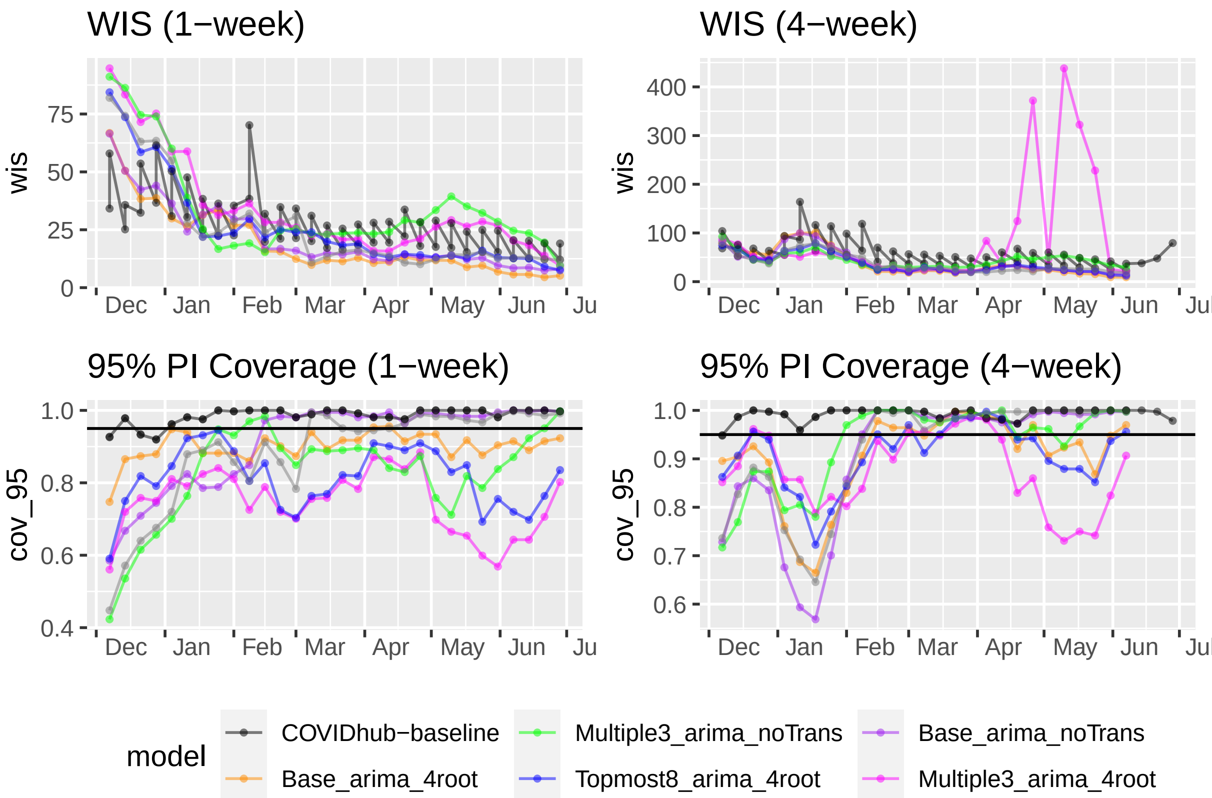


For US National during the alpha wave, the Base_arima_noTrans model performed the best for coverage rates and second best for WIS and MAE at every horizon. The Base_arima_4root model performed best in terms of WIS and MAE but was third for coverage rate while the Topmost8_arima_noTrans model had the second best coverage rates and third best WIS and MAE. During the alpha wave, these models performed the best and tended to have more consistent values for their metrics. The winter21 wave saw noticeably worse values for all metrics at every horizon. Generally, the two models that use the base set hierarchy tended to perform the best for all metrics, although the coverage rates varied a lot by horizon, with the Topmost8_arima_4root and Multiple3_arima_noTrans achieving closer to nominal coverage rates for the 95% interval than even the Base models at horizons greater than 1 week ahead. These results are pretty consistent with improvements over the base-line, which ranged from modest to significant for the best performing models during the alpha wave, especially for WIS and 50% PI coverage. The winter21 wave saw fairly comparable or worse performance compared to the baseline for WIS and MAE. The 50% coverage was better than the baseline for all models, the greatest improvements for the best models, but the 95% coverage could be better or worse for the best models.

For states during both waves, we again see noticeably better coverage rates that approach the nominal level (or even exceed it) for all of the models, compared to those of the US National level when broken into waves. During the alpha wave, the Base_arima_4root model has the best performance for all metrics across all horizons. It is followed closely by the Topmost8_arima_noTrans and Base_arima_noTrans models, then the Topmost8_arima_4root and Multiple3_arima_noTrans models, all of which are much better than the Multiple3_arima_4root model. During the winter21 wave, the Base_arima_4root model is again generally the best for all metrics at all horizon weeks but sometimes loses out in terms of 95% coverage at longer horizons to other models. Notably during this wave, there are much greater similarities in coverage rates and certain models have better coverage rates for the 50% interval while others have better coverage reads for the 95% one. Similar patterns to the ones described above for the US national level when compared to the baseline were observed.

By forecast date (week)





Scoring Conclusions

It seems that the best three models that should be passed onto the testing phase are the Base_arima_noTrans, the Base_arima_4root, and the Topmost8_arima_noTrans models, which seem to take turns being most accurate depending on state or national level overall and during different waves.